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Milestone Three: Algorithms and Data Structures Enhancement

Artifact Description

The artifact I chose for this milestone is my Android application called *DailyDrop*. The purpose of this app is to help users track their daily weight in a simple and consistent manner. The artifact was originally created in CS 360 as a mobile app prototype, and I have since been expanding and refining it for my capstone course. For this milestone, I focused specifically on the algorithms and data structures that support the core functionality of weight tracking, analysis, and reporting. The app now includes enhancements that go beyond storing and displaying raw data, offering meaningful insights to the user such as rolling averages and outlier detection.

Justification of the Artifact

I selected this artifact because it provides a natural opportunity to showcase my skills in algorithms and data structures while building on an application I have steadily improved throughout my program. Weight tracking data lends itself to time-based analysis, which makes it well suited for implementing ordered data structures and algorithms that analyze trends over time. By enhancing *DailyDrop*, I was able to demonstrate my ability to design and integrate a TreeMap and HashMap combination for efficient lookups, updates, and range queries. In addition, I implemented algorithms for calculating seven-day moving averages and for detecting statistical outliers using the median absolute deviation method. These enhancements highlight my ability to apply both theoretical computer science concepts and practical programming techniques to a real application.

Outcomes Achieved

The enhancements I implemented align with the program outcome of designing and evaluating computing solutions using algorithmic principles and appropriate data structures. By creating an in-memory index of weight entries that combines a TreeMap and HashMap, I improved the performance of queries that would otherwise require repeated full scans of the database. This structure supports both chronological ordering and fast updates or deletions, resulting in a balance between efficiency and correctness. The seven-day moving average algorithm provides a meaningful rolling statistic that helps smooth out daily fluctuations, while the outlier detection algorithm flags entries that fall far outside normal variation. Together, these outcomes demonstrate competency in creating innovative techniques that deliver value to end users while also showcasing a professional understanding of algorithm design and complexity trade-offs.

Reflection on the Enhancement

Through the process of enhancing and modifying this artifact, I learned how important it is to consider both usability and efficiency when designing software. Simply storing weights in a database was sufficient for a prototype, but creating a data structure layer that mirrored the database enabled much faster operations and more robust analytics. One of the challenges I faced was ensuring that the in-memory index stayed synchronized with the underlying SQLite database after inserts, updates, or deletions. Solving this problem required careful planning and testing. Another challenge was deciding how to present algorithmic results in a way that would be useful to the user without overwhelming them with technical details. I addressed this by surfacing the seven-day average and outlier count in simple messages and SMS summaries. Overall, these enhancements deepened my appreciation for the role of algorithms and data structures in transforming raw data into meaningful insights for real people.

